April 5, 1938. A. F. DIETZ 2,112,906

ACCELERATING BLASTING PLUG

Filed June 5, 1935

Fig. 1.

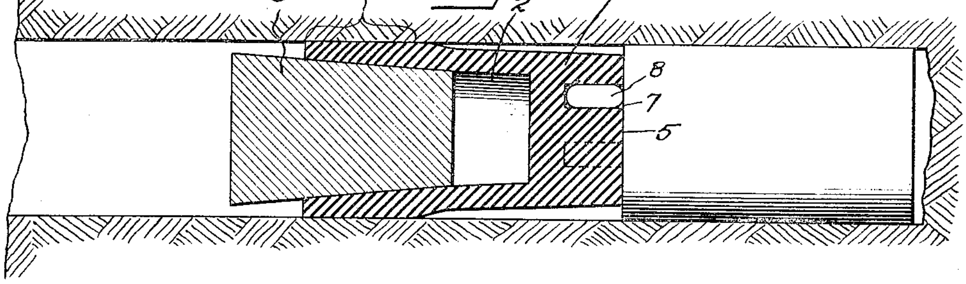


Fig. 2.

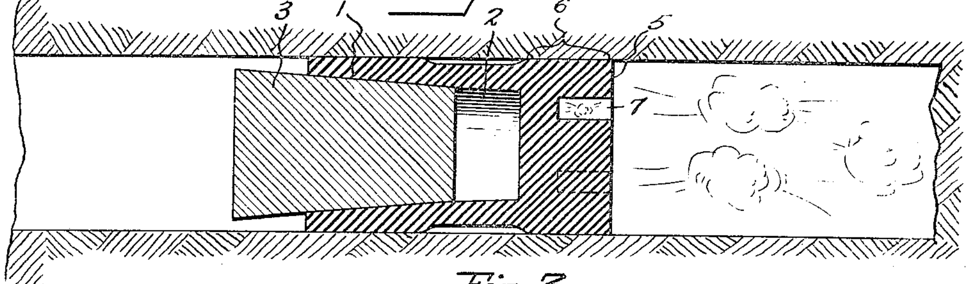


Fig. 3.

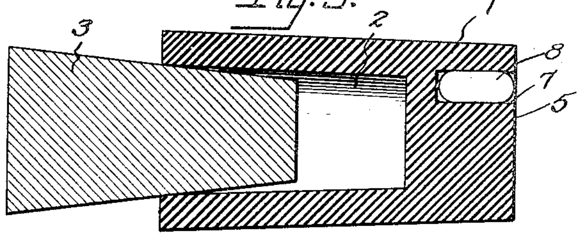


Fig. 5.

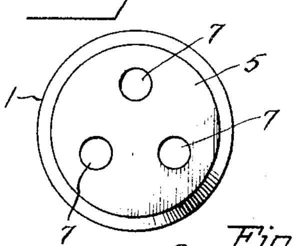


Fig. 4.

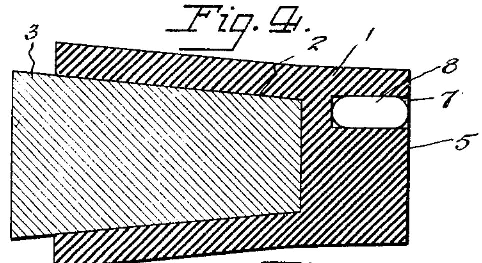


Fig. 7.

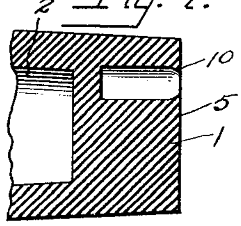
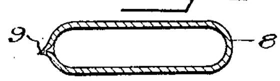


Fig. 6.



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2,112,906

UNITED STATES PATENT OFFICE

2,112,906

ACCELERATING BLASTING PLUG

Alvin F. Dietz, Shamokin, Pa.

Application June 5, 1935, Serial No. 25,158

2 Claims. (Cl. 102—11)

This invention relates to a blasting plug with one or more accelerator chambers and it has for its general object the employment of the blasting plug as the carrier of one or more small vessels containing an accelerator for the explosion and which may itself be an explosive such as nitroglycerin, or preferably, in the interest of safety in storing and use, a non-explosive which is a supporter of combustion, such as liquid oxygen.

The invention preferably utilizes a soft rubber plug of the type disclosed in Patent No. 2,007,568, granted July 9, 1935, to Richard J. Heitzman, which includes a tapered insert by means of which the plug is expanded into sealing relation to the wall of a drilled hole by pressure of a tamping rod, and which at the instant of the explosion is further expanded into sticking contact with the wall of the bore through the explosion pressure, reacting with a cushioning function upon the explosion which reduces the shattering effect and brings down a greater proportion of large pieces of the blasted material.

One of the objects of the invention is to provide one or more cells in the advance end of the plug adapted to contain directly or in suitable vessels, a charge of the accelerating substance.

Another object of the invention is the provision in the inner end of a plug of the class described, of one or more recesses having elastic walls adapted to receive and retain a small vessel or vessels holding the accelerant.

Still another object of the invention is to provide a blasting plug of soft rubber or similar substance having recesses in its inner end adapted to facilitate the bulging of said plug into better sealing contact with the wall of the drilled hole at the instant of explosion, as well as to constitute an air cushion for dampening the shattering effect of the explosion.

Other objects of the invention will appear as the following description of a preferred and practical embodiment thereof proceeds.

In the drawing which accompanies and forms a part of the following specification and throughout the several figures of which the same characters of reference have been employed to designate identical parts:

Figure 1 is an axial section through a drilled hole in rock or coal showing a blasting plug embracing the features of my invention sealing the hole in advance of the explosion charge;

Figure 2 is a similar view showing conditions at the instant of explosion just before disruption of the plug;

Figure 3 is a side sectional view of the blasting plug of my invention in normal or unexpanded condition;

Figure 4 is a similar view showing the insert driven home and the walls of the plug expanded;

Figure 5 is an inner end view;

Figure 6 is a longitudinal section through a capsule adapted to contain liquid oxygen or other accelerant; and

Figure 7 is an axial section through the end of the blasting plug showing the soft rubber retaining flange surrounding the mouth of the capsule receiving recess.

Referring now in detail to the several figures the numeral 1 represents a blasting plug which may be of rubber or any substance having similar qualities of elastic deformation having a recess 2 in its outer end in which is seated a conical insert 3. The insert 3 is preferably of harder substance than the rubber so that as the insert is forced into the recess 2 the wall of the recess 2 will be expanded outward. When this is done in a drilled hole in the coal or rock, the yielding wall of the recess 2 is forced into intimate contact with the wall of the drilled hole along the zone indicated at 4 in Figure 1. This not only seals the explosive from the atmosphere in a perfectly gas-tight manner, but also retains the plug in the hole up to the peak period of the explosion, thus obviating the need of "stemming" which characterizes the customary practice of placing a charge in blasting, particularly in coal mines.

At the instant of explosion the inner end 5 of the plug is forced back, spreading the rubber radially so that the inner end of the plug makes sticking contact with the wall of the drilled hole along the zone 6 in Figure 2. Thus at the instant of explosion the plug is doubly sealed against the escape of gases and held in place long enough to permit the force of the explosion to be extended upon the surrounding coal or rock. The recession of the inner end 5 under the explosion pressure exercises a cushioning effect upon the impact of the explosion, reducing its shattering qualities so that a greater proportion of large pieces of blasted material is thrown down, than is customary with ordinary methods of blasting.

Such plugs are known, being disclosed and claimed in the application for patent to Heitzman aforementioned. My improvement begins with the provision in the anterior end of the plug of one or more bores or recesses 7, preferably of cylindrical form for ease of manufacture. Figure

5 shows that three such recesses are contemplated in the illustrative embodiment of the invention. The number of recesses however is immaterial and may vary from a single recess to as large a number as is practical with respect to the area of the inner end of the plug.

These recesses 7 serve several new and advantageous functions. In the first place, by perforating the mass of rubber in the anterior end of the plug they make it easier for the plug to spread under the explosion pressure extending the zone of sealing contact between the anterior end of the plug and the surrounding wall of the drilled hole. They also constitute air cushions, the inert air in them yielding directly to the explosion pressure.

Thus, although the recesses are primarily intended for the reception of capsules or other suitable vessels of accelerant, they have a decided utility even when the plug is employed without any accelerant. The accelerating substance may be of any substantial nature as has been indicated, for example, it may be a charge of nitroglycerin. This might be poured directly into the recesses 7 and said recesses corked before the plug is thrust home in the drilled hole. The softness and yielding characteristic of which the plug is made would preclude the premature uniting of the nitroglycerin through tamping pressure necessary to seat the plug. Preferably the nitroglycerin would already be packed in small ampuls or capsules such as the capsule 8 shown in Figure 3 and one or more of these capsules may be inserted in the recesses at the time of placing the charge. However, nitroglycerin at best is an uncertain and unsafe substance to handle and it is preferred to use an accelerant which is in itself inexplosive, but which will support combustion and in that way add to the power of the explosion. Oxygen is suggested as a safe accelerant of this nature and in order to get enough of it into the drilled hole it would have to be supplied in liquid form. Figure 6 shows a fluid-tight capsule 8 which may be glass and containing liquid oxygen. The ampul 8 is shown as being blown and filled like an electric light bulb with a fused teat 9 on the end. By approved methods of manufacture, this teat could be omitted and the capsule be of uniform contour throughout. In place of the capsule, an ordinary small bottle with sealed stopper may serve as the container of the oxygen. Regardless of the type of container of accelerant, the recesses 7 are preferably so shaped as to fit and retain the capsule. Since the walls of the recesses 7 are of yielding elastic, the recesses may, if desired, be made slightly smaller than the capsule and the latter forced into the recess being retained by frictional pressure. Or, as shown in Figure 7 the mouths of the recesses may be provided with a surrounding yielding integral flange 10 of rubber through which the capsule may be pressed into said recess, and after it has entered said recess the flange 10 will spring out in front of the capsule and keep it from being displaced. In use, the explosion is produced in the customary manner and immediately shatters the capsule freeing the oxygen which accelerates the power of the explosion.

The plugs can be made and sold ready loaded with the capsules or a supply of such capsules may be kept on hand and the plugs filled with the charged vessels before being inserted into the drilled hole.

Although I have illustrated my invention with a somewhat specific form of the invention, it will be understood to those skilled in the art that the details of construction are merely by way of example and not to be construed as limiting the scope of the appended claims.

What I claim is:

1. Blasting plug comprising a cylindrical member of a material having the quality of elastic deformation comparable with that of soft rubber, having a recess at its outer end and an expander in said recess for expanding the periphery of said member into sealing relation to a drilled hole in coal or rock, the inner end of said member being deformable under pressure to form an auxiliary peripheral seal to cushion the impact of the explosion, said inner end being formed with one or more flexible walled chambers opening at the face of the inner end and a normally sealed vessel containing an explosion accelerant for the said charge, said accelerant being liberated from said vessel by the shock of the explosion.

2. Blasting plug comprising a cylindrical member of a material having the quality of elastic deformation comparable with that of soft rubber, having a recess at its outer end and an expander in said recess for expanding the peripheral wall of said member into sealing relation to the drilled hole in coal or rock, the inner end of said member being deformable under pressure to form an auxiliary peripheral seal and to cushion the impact of the explosion, said inner end being formed with one or more chambers opening at the face of said inner end, and one or more vessels of an explosion accelerant positioned in said chambers, the mouths of said chambers being provided with integral flexible flanges functioning as check valves permitting the insertion of said vessels into said chambers, but preventing their free escape therefrom.

ALVIN F. DIETZ.